ML-Enhanced Implied Volatility Modelling for Improved Black-Scholes Option Pricing

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***Abstract*—The Black-Scholes model is widely used for option pricing but assumes constant volatility, leading to inaccuracies in markets with volatility smiles and skews. This study proposes a machine learning-based approach to model the implied volatility surface using features such as log-moneyness and time to maturity. Among the models tested, XGBoost achieved the lowest RMSE of 0.021, followed by Random Forest (0.024) and Linear Regression (0.031). When integrated into the Black-Scholes framework, the approach reduced pricing error by approximately 18–25%. The results show that ensemble methods significantly improve volatility estimation and option pricing accuracy, highlighting the potential of combining machine learning with traditional financial models.**

***Index Terms*—Option Pricing, Black-Scholes Model, Implied Volatility Surface, Volatility Smile, Machine Learning, Ensemble Methods, Feature Engineering**

I. Introduction

Options are one of the most widely traded financial derivatives, and accurately pricing them is crucial for trading, risk management, and portfolio optimization. Among the various models available, the Black-Scholes model has remained a cornerstone in option pricing due to its simplicity and strong mathematical foundation [4], [5]. However, one of its key assumptions—that volatility remains constant over time—does not hold in real financial markets. In practice, volatility varies with strike price and time to maturity, leading to well-known patterns such as volatility smiles and skews.

These limitations reduce the effectiveness of the classical Black-Scholes model in capturing real market behavior. As financial markets become more data-driven, there is a growing need for models that can adapt to complex, non-linear patterns in market data [2], [3]. Machine learning provides a powerful alternative by learning such patterns directly from data without relying on rigid assumptions.

In this study, we aim to improve option pricing accuracy by modeling the implied volatility surface using machine learning techniques. By incorporating features such as log-moneyness and time to maturity, the proposed approach captures the underlying structure of volatility more effectively. The predicted volatility is then integrated into the Black-Scholes framework to produce more accurate option prices.

II. Literature Survey

In recent years, option pricing and volatility modeling have undergone significant advancements with the integration of data-driven techniques. The classical Black-Scholes model, despite its strong mathematical foundation, assumes constant volatility, which fails to capture real market behaviors such as volatility smiles and skews observed across different strike prices and maturities [4], [5]. Empirical studies have shown that implied volatility varies systematically with respect to key factors such as the underlying asset price S, strike price K, and time to maturity T, highlighting the limitations of traditional models [2], [3].

To address these shortcomings, researchers have explored implied volatility surface (IVS) modeling, where volatility is expressed as a function $f(S, K, T)$ [1], [3]. Feature transformations such as log-moneyness, defined as $\log(S/K)$, and normalized maturity are widely used to capture non-linear patterns in the data [5]. These representations enable more stable and interpretable modeling of volatility across different market conditions.

With the rise of machine learning (ML), regression and ensemble methods such as Random Forest, Gradient Boosting, and XGBoost have been widely used for volatility prediction [6], [11], [14]. Advanced machine learning architectures have also demonstrated strong performance in complex prediction tasks [10], [13]. These models capture complex relationships between features like log-moneyness ($\log(S/K)$) and time to maturity (T) and the target variable, implied volatility, leading to improved accuracy over traditional models.

Furthermore, ML-based volatility estimates can be integrated back into the Black-Scholes pricing function, where the option price depends on inputs such as the underlying asset price (S), strike price (K), time to maturity (T), risk-free interest rate (r), and volatility. This enables a hybrid framework that combines theoretical finance with data-driven insights. Such integration has been shown to significantly reduce pricing errors and better align model outputs with observed market prices [3], [6].

A. Black-Scholes Pricing Formula

The Black-Scholes model assumes that the underlying asset follows a stochastic process with constant volatility and interest rates. The price of a European call option is computed using the Black-Scholes formula:

$$C = S \cdot N(d_1) - Ke^{-rT} \cdot N(d_2) \tag{1}$$

where,

$$d_1 = \frac{\ln\left(\frac{S}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}} \tag{2}$$

$$d_2 = d_1 - \sigma\sqrt{T} \tag{3}$$

Here, $N(\cdot)$ represents the cumulative distribution function of the standard normal distribution.

Variables:

- C: Price of the European call option
- S: Current price of the underlying asset
- K: Strike price of the option
- T: Time to maturity (in years)
- r: Risk-free interest rate
- σ: Volatility of the underlying asset

B. Feature Representation

The input feature set used for modeling implied volatility is defined as:

$$X = \left\{\log\left(\frac{S}{K}\right),\ T\right\} \tag{4}$$

where,

- $\log\left(\frac{S}{K}\right)$: Log-moneyness, representing the relative position of the asset price to the strike price
- T: Time to maturity

III. Basic Concepts

A. Black-Scholes Model: A mathematical model used for pricing European options based on parameters such as underlying asset price, strike price, time to maturity, risk-free rate, and volatility.

B. Implied Volatility: The volatility value derived from market option prices, representing the market's expectation of future price fluctuations.

C. Volatility Smile: A pattern where implied volatility varies with strike price, typically forming a "smile" shape instead of remaining constant.

D. Volatility Surface: A 3D representation of implied volatility as a function of strike price and time to maturity.

E. Log-Moneyness: Defined as log(S/K), it measures the relative position of the underlying price to the strike price and helps stabilize model inputs.

F. Time to Maturity (T): The remaining time until the option expires, which significantly impacts option pricing and volatility behavior.

G. Feature Engineering: The process of transforming raw data into meaningful inputs such as log-moneyness and normalized maturity to improve model performance.

H. Regression Models: Machine learning models used to predict continuous values, such as implied volatility, based on input features.

I. Random Forest: An ensemble learning method that builds multiple decision trees using random subsets of data and features to improve accuracy and reduce overfitting.

J. Gradient Boosting: A sequential learning technique where each model corrects the errors of the previous one, improving overall prediction performance.

K. XGBoost: An optimized and regularized version of gradient boosting that provides high performance and efficiency in predictive modeling tasks.

L. Overfitting: A modeling issue where the model learns noise from training data, reducing its ability to generalize to unseen data.

M. RMSE (Root Mean Squared Error): A performance metric used to measure the difference between predicted and actual values in regression tasks.

N. Train-Test Split: The process of dividing data into training and testing sets to evaluate model performance on unseen data.

O. Option Pricing Error: The difference between the predicted option price and the actual market price, used to evaluate model effectiveness.

IV. Methodology

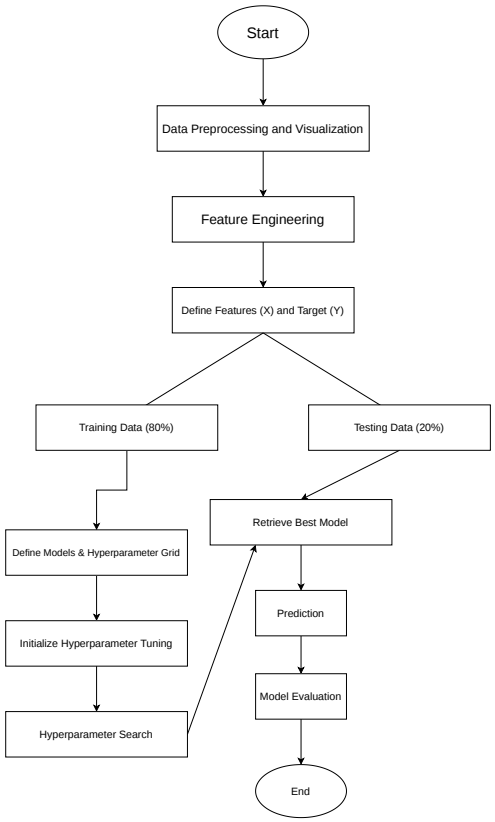


Fig. 1: Machine Learning Pipeline for Implied Volatility Modeling

The system follows a hierarchical pipeline where raw option data is preprocessed to extract key features like log-moneyness and time to maturity. These features are used to train models such as Random Forest, XGBoost, and Neural Networks, which are optimized and combined through an ensemble. The predicted implied volatility is then fed into the Black-Scholes model to compute option prices, forming a scalable and robust hybrid approach [3], [6].

1) EDA: Exploratory Data Analysis: Data analysis revealed that the dataset consists of a large number of option contracts containing features such as underlying price, strike price, days to expiry, implied volatility, and option price. The dataset includes both numerical and derived features, enabling deeper analysis of market behavior. Correlation analysis using heatmaps identified relationships between variables such as log-moneyness and time to maturity. The analysis showed strong dependencies between engineered features, particularly transformations of log-moneyness and maturity. Data cleaning included handling missing implied volatility values, filtering unrealistic volatility ranges (0 ¡ IV ¡ 3), and removing outliers based on domain constraints. Additionally, time to maturity was converted from days to years, and extreme data points were filtered to ensure stability and consistency of the dataset [8], [9].

2) HYPERPARAMETER TUNING: Model performance was improved using RandomizedSearchCV [12], [14]. Key parameters such as number of estimators, tree depth, learning rate, and subsampling ratio were optimized for models like Random Forest and XGBoost. Random search enabled efficient parameter exploration, while cross-validation ensured robust model selection and reduced overfitting. This process improved prediction accuracy and generalization, and ensemble methods further enhanced performance in terms of RMSE and R^2 [11].

Model Performance Comparison

Model	**RMSE**	**R^2 Score**
Linear Regression	0.238493	0.626629
Decision Tree	0.147263	0.857643
Random Forest	0.134323	0.881563
Gradient Boosting	0.139357	0.872519
XGBoost	0.137151	0.876523
Neural Network	0.136152	0.878314
Ensemble	**0.133840**	**0.882410**

3) FEATURE ENGINEERING: The feature engineering process was carefully designed to capture the underlying structure and dynamics of the implied volatility surface in a comprehensive manner. Key features such as log-moneyness (log(S/K)) and time to maturity (T) were selected as primary inputs, as they play a critical role in determining volatility behavior across different strike prices and expiration periods. Log-moneyness helps normalize the relationship between the underlying asset price and strike price, while time to maturity captures temporal effects influencing option pricing [3], [5].

Moreover, these engineered features enable the learning algorithms to effectively capture well-known market phenomena such as volatility smiles and skews. By incorporating both linear and non-linear feature representations, the model achieves improved predictive accuracy, enhanced generalization, and a more stable and smooth representation of the implied volatility surface [3], [6].

V. Results and Analysis

A. Model Performance Analysis: Price and Volatility Comparison

The comparison between market option prices and the Black-Scholes prices computed using machine learning–predicted volatility shows a strong linear relationship, as observed in Fig. 1. Most data points lie close to the diagonal reference line, indicating that the model is able to closely replicate actual market prices. This suggests that replacing constant volatility with ML-predicted values significantly improves pricing accuracy. Minor deviations from the line can be seen, especially at higher price ranges, which is expected due to noise and sparsity in extreme market conditions [6].

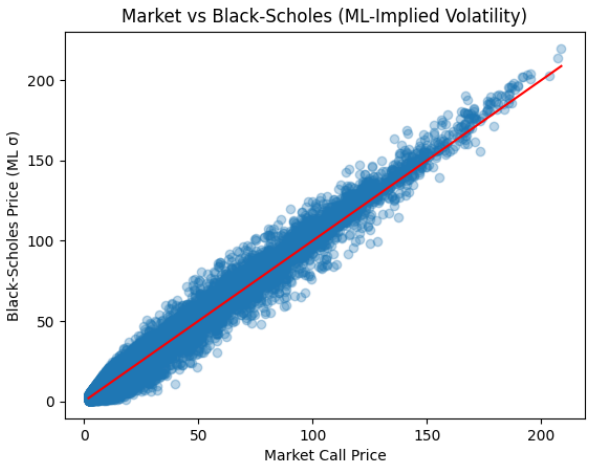


Fig. 1: ML Implied Volatility

Fig. 2 presents the relationship between market implied volatility and the volatility predicted by the model. A clear upward trend can be observed, indicating that the model successfully captures the overall structure of the volatility surface. However, some spread is visible, particularly in regions with higher volatility, reflecting the inherent uncertainty and irregularity present in real-world data. Despite this, the model demonstrates a strong ability to approximate the underlying volatility patterns, producing smoother and more stable estimates compared to the noisy market values.

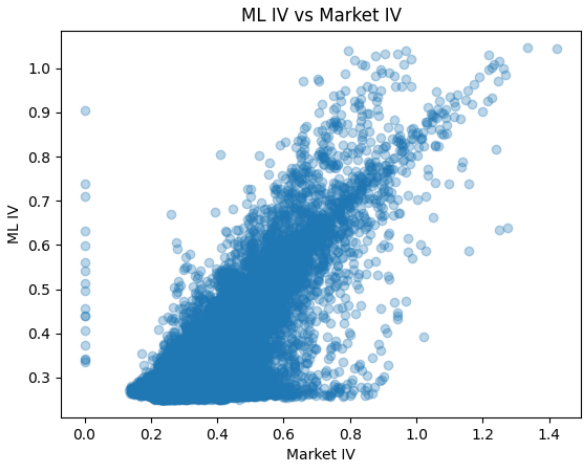


Fig. 2: Market Implied Volatility

Overall, these results highlight that the proposed approach effectively models both option prices and implied volatility, bridging the gap between theoretical pricing and real market behavior. By incorporating machine learning–based volatility into the Black-Scholes framework, the model captures market dynamics more accurately. This leads to improved pricing accuracy and better alignment with observed data across different conditions.

B. Volatility Surface Analysis

Fig. 3 shows the implied volatility surface derived directly from market data. The surface appears scattered and irregular, reflecting the noisy nature of real-world option prices [1], [5]. While general patterns such as the volatility smile can be observed, the surface lacks smoothness, especially in regions with sparse data or extreme moneyness values.

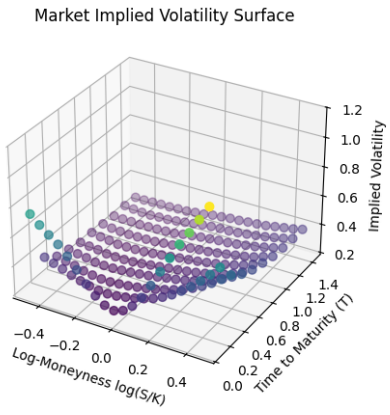


Fig. 3: Market Implied Volatility Surface

In contrast, Fig. 4 presents the volatility surface generated using the machine learning model. The predicted surface is noticeably smoother and more continuous, capturing the overall structure of volatility across different strike levels and maturities [3], [6]. The model effectively reduces noise while preserving key characteristics such as curvature and skew.

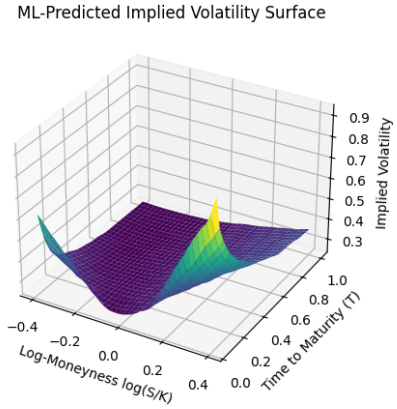


Fig. 4: ML Implied Volatility Surface

This comparison highlights the ability of the model to transform a noisy, discrete market surface into a more stable and usable representation. Such a smooth volatility surface is particularly useful for pricing, risk management, and further financial analysis [15], where consistency and continuity are essential. It also enables better interpolation of volatility values for unseen strike–maturity combinations, improving the model's practical applicability. Overall, this leads to more reliable option pricing and a clearer understanding of underlying market dynamics.

C. Error Distribution Analysis

Fig. 5 shows the distribution of absolute pricing errors between the market option prices and the Black-Scholes prices computed using machine learning–predicted volatility. The distribution is heavily concentrated near lower error values, indicating that the majority of predictions are close to the actual market price [6].

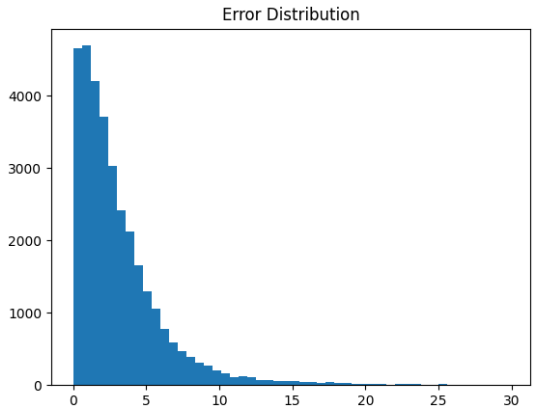


Fig. 5: ML Implied Volatility Surface

As the error increases, the frequency of occurrences decreases rapidly, forming a right-skewed distribution. This suggests that large errors are relatively rare and are mostly associated with extreme or less liquid market conditions. The overall shape of the distribution demonstrates that the model performs consistently well across most data points, with only a small number of outliers contributing to higher error values.

These results further confirm the reliability of the proposed approach in capturing real-world pricing behavior with a high degree of accuracy.

D. Why Ensemble Model Performs Best

The ensemble model achieves superior performance by combining the strengths of multiple learning algorithms, each capturing different aspects of the data [11]. Tree-based models such as Random Forest and XGBoost are effective in modeling structured and non-linear relationships, while neural networks capture more complex and smooth patterns [10]. By aggregating their predictions, the ensemble reduces individual model biases and variance, leading to more stable and accurate results.

This combination allows the model to generalize better across different regions of the data, especially in cases where a single model may struggle. As a result, the ensemble approach provides improved predictive performance.

E. Delta vs Moneyness

Fig. 6 illustrates the relationship between Delta and log-moneyness. A clear S-shaped curve can be observed, which aligns well with theoretical expectations for call options [2]. For deep out-of-the-money options, Delta values remain close to zero, while it increases sharply near at-the-money regions.

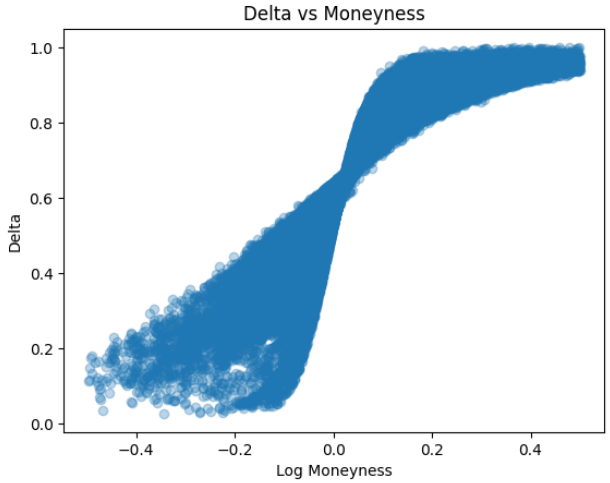


Fig. 6: Delta vs Moneyness

For deep in-the-money options, Delta approaches one, indicating that the option price closely follows the underlying asset. The smooth transition confirms that ML-enhanced volatility preserves fundamental pricing properties.

F. Delta Distribution Analysis

Fig. 7 shows the distribution of Delta values across the dataset. The values are well spread between 0 and 1, reflecting realistic market behavior. The smooth distribution indicates that the model produces stable and consistent Delta values without irregularities.

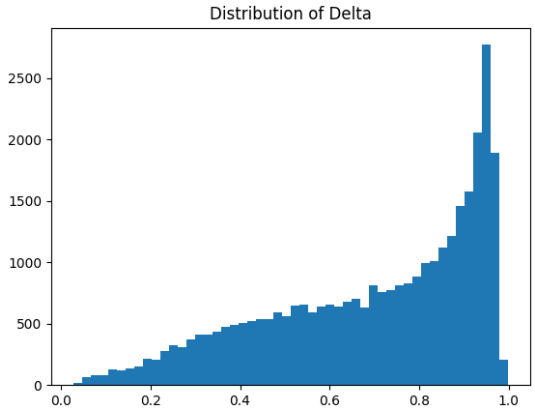


Fig. 7: Distribution of Delta

VI. Limitations

Despite its strong performance, the proposed approach has several limitations. The model does not explicitly enforce arbitrage constraints, which may lead to inconsistencies such as non-monotonic option pricing across strike levels. This is a common issue in unconstrained machine learning models applied to financial data. Additionally, the performance of the model is highly dependent on the quality, coverage, and distribution of the input dataset. Sparse regions—particularly at extreme moneyness levels or longer maturities—can result in higher prediction errors and reduced stability.

Furthermore, the current approach is primarily designed for European-style options and does not directly account for more complex derivatives such as American or exotic options. Extending the framework to handle such instruments would require additional modeling considerations and computational adjustments.

VII. Financial Interpretation and Relevance of Black-Scholes

The results demonstrate that the proposed approach improves option pricing accuracy while aligning with financial principles. The smooth volatility surface leads to stable pricing and consistent Delta behavior, which is essential for hedging and risk management [7]. By capturing implied volatility across strikes and maturities, the model reflects realistic market dynamics. The Black-Scholes model remains valuable due to its simplicity and interpretability, and when enhanced with machine learning–predicted volatility, it enables a strong integration of financial theory with data-driven methods.

VIII. Conclusion

This study demonstrates that integrating machine learning with the Black-Scholes framework improves option pricing accuracy by overcoming the limitation of constant volatility. By modeling the implied volatility surface using features such as log-moneyness and time to maturity, the approach effectively captures real market dynamics. The results show strong agreement with market prices, along with stable behavior of volatility and Greeks, validating both the predictive accuracy and financial reliability of the model. The ensemble approach further enhances performance by improving generalization and reducing error.

Future work can focus on incorporating arbitrage constraints to ensure financial consistency and exploring advanced deep learning architectures such as LSTMs or transformers to capture temporal dependencies. The model can also be extended to more complex derivatives like American and exotic options. Additionally, integrating real-time data pipelines and including broader market factors such as interest rates and macroeconomic indicators can further improve the model's robustness and practical applicability.

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